



Peripheral oxytocin receptors inhibit the nociceptive input signal to spinal dorsal horn wide-dynamic-range neurons

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Abstract

Oxytocin (OT) has emerged as a mediator of endogenous analgesia in behavioral and electrophysiological experiments. In fact, OT receptors (OTRs) in the spinal dorsal horn participate in a selective inhibition of the neuronal activity mediated by A δ and C fibers but not A β fibers. This study shows that OTRs are expressed in the terminal nerve endings and are able to inhibit nociceptive neuronal firing. Indeed, local peripheral OT blocked the first sensorial activity of A δ and C fibers recorded in the spinal cord neurons. Furthermore, using the formalin behavioral nociceptive test, we demonstrated that only ipsilateral OTR activation inhibits pain behavior. Our data are reinforced by the fact that the OTR protein is expressed in the sciatic nerve. Consistent with this, immunofluorescence of primary afferent fibers suggest that OTRs could be located in nociceptive-specific terminals of the skin. Taken together, our results suggest that OTRs could be found in nociceptive terminals and that on activation they are able to inhibit nociceptive input.

Keywords: Analgesia, C fibers, Oxytocin receptor, Pain, WDR cells

1. Introduction

Oxytocin (OT) is a nonapeptide that is released mainly into the central nervous system by the hypothalamic paraventricular and supraoptic nuclei.⁴⁶ Oxytocin binding sites and receptors are widely extended.¹⁸ This peptide is involved in different physiological functions such as: (1) cardiovascular activity; (2) breathing; (3) feeding; (4) social interactions; and more recently, (5) nociception.^{2,7,33,41} Regarding nociception, OT has emerged as an interesting molecule to induce analgesia at the spinal cord level, not only in animal pain models^{12,16,37,42} but also in humans.^{9,10} More recently, it was described¹⁶ as a particular cell population in the parvocellular part of the paraventricular nuclei that controls magnocellular OT activity and suppress nociception under activation. In fact, Juif and Poisbeau²⁶ demonstrated that intravenous OT administration diminishes the evoked neuronal activity associated with C-fiber activation in wide-dynamic-range (WDR) cells pointing out that this effect is at the dorsal root ganglion/DRG level. More recently, it has been suggested that OT

released from the supraoptic nuclei modulates nociception by activating oxytocin receptors (OTRs) located in DRG nociceptive neurons through a similar mechanism.¹⁶ Furthermore, experiments using whole-cell patch clamps and voltage clamps on isolated DGR neurons have found that OT inhibits the acid-sensing ion channel currents induced by acidification,⁴³ and there are similar results from studies using capsaicin.²² Certainly, OT does not cross the blood-brain barrier,¹⁷ and the current consensus about the peripheral antinociceptive properties of OT is restricted at the DRG level. Because OTRs are expressed in DRG cells^{40,55} and in the trigeminal ganglia,⁴⁹ no study has yet reported on its role in the peripheral terminal nociceptive nerve endings. However, these studies have failed to extend its potential role of peripheral action of OT on terminal nerve endings (ie, on superficial skin layer) as observed for other antinociceptive systems.³

In this context, the present study was designed to describe and test not only the potential peripheral local antinociceptive action of OT, but also the functional role of OTR in peripheral nociceptive skin terminals. To conduct this study, we used electrophysiological (extracellular unitary recordings of spinal dorsal horn WDR cells), a behavioral nociceptive test (formalin test), and molecular and pharmacological tools to reveal a new mechanism at the periphery modulating nociception. We present evidence for a specific role of OT in OTR modulating the nociceptive input at the terminal nerve endings.

2. Experimental procedures

2.1. Animals

A total of 121 male Wistar rats (220–280 g) from the Neurobiology Institute Animal House were used in the experiments. Animals

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

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Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.painjournalonline.com).

PAIN 158 (2017) 2117–2128

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<http://dx.doi.org/10.1097/j.pain.0000000000001024>

were maintained on a 12:12 hours light and dark cycle (light beginning at 07:00 hours) and housed in a special room at constant temperature ($22^{\circ}\text{C} \pm 2^{\circ}\text{C}$) and humidity (50%) with food and water ad libitum. Our Institutional Ethics Committee approved all animal procedures and protocols, and they followed the IASP ethical guidelines,⁵⁸ a Guide for the Care and Use of Laboratory Animals established by the NIH, and ARRIVE guidelines for reporting experiments involving animals.³⁵ All efforts were made to limit distress and use only the number of animals necessary to produce reliable scientific data.

2.2. General methods

2.2.1. Surgical procedures for electrophysiological experiments

Animals were anesthetized with urethane (1–1.2 g/kg) and then an intratracheal cannula was inserted for artificial ventilation (55 strokes/min). Subsequently, animals were mounted onto a stereotaxic frame and secured in a spinal cord unit frame; the lumbar vertebrae were fixed to improve stability at the recording site to perform a laminectomy to expose the L2–L4 spinal cord segments. The dura was carefully removed, and to avoid desiccation, we covered the exposed spinal cord with mineral oil. The animals were not paralyzed, and we did not observe a withdrawal reaction during the experiments as previously reported.³⁶ End tidal CO_2 was monitored with a Capstar-100 CO_2 analyzer (CWI Inc, Ardmore, PA) and kept between 2.5% and 3.0% by adjusting the stroke volume to maintain a normal acid-base equilibrium. Core body temperature was maintained at 38°C using a circulating water pad.

2.2.2. Extracellular unitary recordings

Extracellular unitary recordings were made using 7 quartz-platinum/tungsten microelectrodes (impedance 4–7 M Ω) mounted in a multichannel microdrive “System Eckhorn.” The multi-electrode system was manipulated with the 7-channel version of the fiber-electrode manipulator “System Eckhorn” using Eckhorn Matrix multiuser software (Thomas RECORDING GmbH, Giessen, Germany). The microelectrodes were lowered (400–900 μm from the surface) in small steps (2–5 μm) into the superficial laminae of the left dorsal horn segments to search for single-unit discharges. For each recorded cell, the specific somatic receptive field (RF) was located by tapping on the entire ipsilateral hind paw and toes; electrical stimulation was then applied by 2 electrodes inserted into the RF (see below). In this case, 2 fine needles (27 G) attached to a stimulus isolator unit were inserted subcutaneously (s.c.) into the RF of the recorded neuron. The electrical test stimulation was then conducted; this test consists of 20 stimuli at 0.5 Hz with 1-millisecond pulse duration at 1.5 times the threshold intensity (0.1–3 mA) required to evoke a C-fiber response.

The extracellular neuronal activity induced by electrical stimulation of the RF was recorded, amplified, digitalized, and discriminated using CED hardware and Spike 2 software (Version 5.20; Cambridge Electronic Design, Cambridge, United Kingdom). Raw and discriminated signals were fed through an audio monitor and displayed on an oscilloscope. Waveforms and recorded spike trains were stored on a hard drive for off-line analysis. Baselines and evoked activities of the spinal dorsal horn WDR neurons were recorded and analyzed as cumulative frequency and post-stimulus time histograms to detect the occurrence of statistically significant neuronal responses. All WDR cells recorded were found between 450 to 750 μm from the

surface. On this basis, the stimulating threshold to evoke action potentials and their frequency of occurrence, resulting from the stimulation of the peripheral RF on the hind paw, were attributed to the recruitment of A β -, A δ -, and C fibers. Considering the distance between the RF and the recording electrode, the peak latencies observed correspond to peripheral conduction velocities⁵⁰ within the A β - (0–20 milliseconds), A δ - (21–90 milliseconds), and C fibers (90–350 milliseconds) (Fig. 1A–D). Thus, the number of action potentials that occurred in response to 20 RF stimuli was compared before (basal) and after vehicle/drug (50 μL ; s.c., in the RF) treatment.

Accordingly, the neuronal-evoked responses were evaluated immediately after ($t = 0$) OT (1, 10 and 56 $\mu\text{g/paw}$; $n = 6$ cells, each dose) or vehicle (isotonic saline; $n = 5$ cells) administration at 5, 10, 20, 40, 60, 80, and 100 minutes. The role of OTR in the OT effect was also evaluated by s.c. administration of the potent ($\text{pK}_i = 8.1$)²⁵ and selective antagonist⁵⁰ L-368,899 (10 $\mu\text{g/paw}$; $n = 4$ cells). The antagonist was administered 5 minutes before OT or vehicle.

2.2.3. Formalin-induced acute nociception

Acute nociception was assessed using the 1% formalin test.¹⁵ Rats were placed in open Plexiglass observation chambers for 1 hour during 3 consecutive days to allow them to become familiar with their surroundings. On the fourth day, and after 30 minutes in the Plexiglass chamber, they were removed for formalin administration. Formalin (1%) was injected s.c. (50 μL) into the dorsal surface of the right hind paw using a 30-gauge needle. Animals were then returned to the chambers and nocifensive behavior was observed immediately after formalin injection. Nocifensive behavior was quantified as the number of flinches of the injected paws during 1-minute periods every 5 minutes for up to 60 minutes after injection.^{34,52} Flinching was characterized as a rapid and brief withdrawal or flexing of the injected paw. Formalin-induced flinching behavior was biphasic. The initial acute phase (0–10 minutes) was followed by a relatively short quiescent period, which was then followed by a prolonged persistent response (15–60 minutes). At the end of the experiments, the rats were killed in a CO_2 chamber. Oxytocin (0.1, 10, 31, and 100 $\mu\text{g/paw}$; $n = 6$ rats, each dose) or vehicle was administered 5 minutes before formalin. Furthermore, the L-368,899 was given (10 and 100 $\mu\text{g/paw}$; $n = 5$ rats each dose) 5 minutes before OT or vehicle.

2.2.4. Motor coordination test

A motor coordination test (rota-rod test) in an independent group of rats was performed in a treadmill apparatus (IITC Inc Life Science, Los Angeles, CA). Briefly, this test consists of placing the animals on a cylinder (diameter: 7 cm) rotating at a constant speed of 15 revolutions per minute (rpm). Animals were trained to walk on the cylinder (cutoff time: 180 seconds) in 5 previous consecutive sessions without any treatment, and in the sixth session they received 31 $\mu\text{g/paw}$ of OT ($n = 5$) or vehicle (50 μL ; $n = 5$). The time it took to fall was counted.

2.2.5. Immunofluorescence of primary afferent fibers

The left sciatic nerve was exposed in 6 male Wistar rats (280–310 g). Distal to the 3 peripheral branches (sural, common peroneal, and tibial nerves) of the sciatic nerve, the tracer True Blue (TB) was placed under microscopic control; in this case, we used a stainless steel wire charging tiny pellets of TB attached to

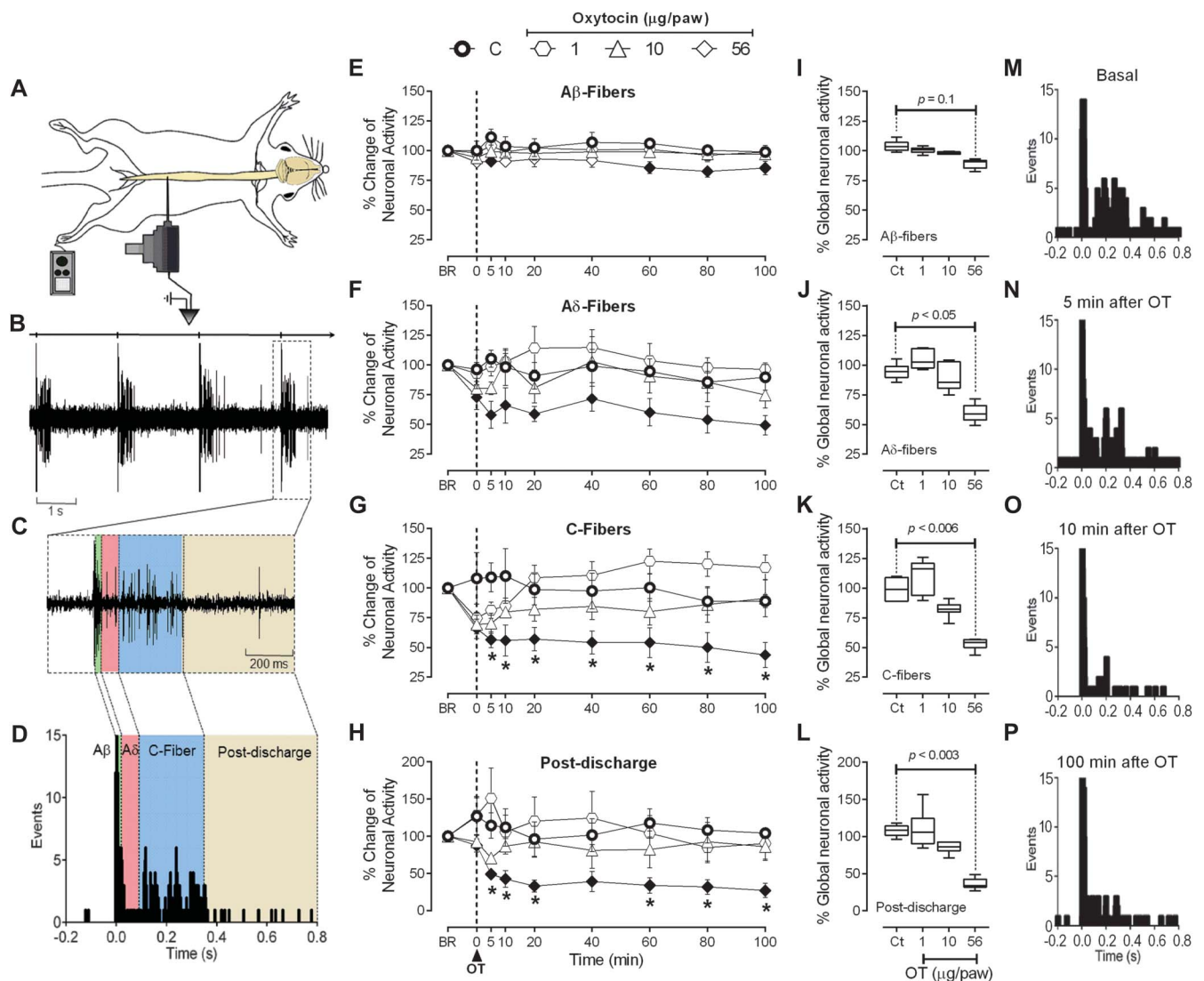


Figure 1. Peripheral oxytocin inhibits the nociceptive activation fibers arriving at second-order wide-dynamic-range (WDR) cells. (A) Experimental setup design illustrating the electrophysiological recording of lumbar spinal dorsal horn WDR cells and the location of the receptive field (RF) stimulation on the paw. (B) Raw data of 4 stimulus artifacts (upper line) and consecutive WDR responses to RF stimulation. (C) Raw tracing of a single response to RF stimulation. (D) Peri-stimulus time histograms (PSTHs) constructed from 20 WDR responses to RF stimulation depicting the different fiber components (A β fibers, A δ fibers, C fibers, and postdischarge). (E, F, G, and H) Time course changes in the percentage average of the different fibers activating WDR cell responses induced by RF stimulation and the effects of subcutaneous (s.c.) oxytocin (OT) injection (1, 10, and 56 μ g/paw; $n = 6$ each dose) at time 0. * $P < 0.05$, statistically significant difference compared at the same time with control (●; $n = 5$) responses. For the sake of clarity, solid symbols (●, ▲, ◆), instead of empty symbols, represent significant ($P < 0.05$) responses vs its respective (100%) initial basal response (BR). Notice that OT predominantly inhibits (compared with control curve, ●) the response of C fibers and postdischarge. Panels I, J, K, and L show global neuronal activity (obtained from the respective time course figures) of A β fibers, A δ fibers, C fibers, and postdischarge in response to s.c. OT; this neuropeptide was able to preferentially block the neuronal activity associated to A δ fibers, C fibers, and postdischarge but not A β fibers. This inhibition is clearly depicted in panels M, N, O, and P, where the PSTHs were obtained for 1 WDR neuron illustrating the effect of s.c. OT (56 μ g/paw) at different times after OT administration. In this case, note that OT blocks the activity associated with the activation of nociceptive fibers (ie, A δ fibers and C fibers).

its blunt tip. After 5 days, the animals were killed by decapitation and immediately after, the lateral hairy skin areas were excised with a scalpel, snap frozen in Tissue-Tek O.C.T. Compound (Sakura Finetek USA, Inc, Torrance, California), placed in liquid nitrogen-chilled isopentane, and stored at -80°C before sectioning.

Triple immunofluorescence for the OTR, IB4, and calcitonin gene-related peptide (CGRP) was performed on free-floating and slide-mounted samples of 50- μ m thick cryostat sections after fixation in 4% phosphate-buffered paraformaldehyde for 10 minutes at room temperature. A primary antibody cocktail was used for 24 hours at 4°C ; it included polyclonal antibodies against OTR (raised in goat, diluted 1:400, Cat.no. sc-8103; Santa Cruz

Biotechnology Inc, Santa Cruz, CA) and the CGRP (raised in rabbit, diluted 1:2000, Cat.no. AB15360; Chemicon International, Temecula, CA). The sections were washed for 5 minutes with 0.1 M PBS and then incubated with the appropriate secondary antibody for 24 hours at 4°C . The secondary antibodies used were donkey anti-Goat IgG (Alexa Fluor 488 conjugate, Cat.no. A-11055; Invitrogen, Grand Island, NY) and donkey anti-Rabbit IgG (Alexa Fluor 647 conjugate, Cat.no. A-31573; Invitrogen). To detect isolectin B4 (IB4) binding, we included 1:400 *Griffonia simplicifolia* isolectin (GS-IB4) Alexa Fluor 568 (Cat.no. I21412; Invitrogen) during secondary antibody incubations.

An extra group of 3 rats was required during the submission process. This group had the same experimental protocol but

the tissue was cut serially in 20 μm -thick cryostat sections. One section was treated as previous experiments and the following sections without the primary antibody for each OTR (1:400), IB4 (1:400), or CGRP (1:2000). This maneuver allowed us to reveal the possible unspecific immunofluorescence for the OTR, IB4, and CGRP in order to avoid a possible false positive staining.

2.2.6. Confocal microscopy and image analysis

Confocal images of the peripheral terminal sensory in the lateral hairy skin were acquired using the LSM510 or LSM780 confocal microscope system (Zeiss, Mexico) with $25\times/0.8$, $40\times/1.3$, and $63\times/1.32$ NA oil immersion objective. Using the 488-nm argon laser to excite Alexa Fluor 488, 561-nm diode laser for Alexa Fluor 568, and 633-nm helium/neon laser to excite the Alexa Fluor 633 signal. The pinhole, Z-sectioning intervals were kept constant for all images. About 40 to 100 optical Z-sections of 1- μm thickness were obtained from the 50- μm thick tissue for each skin image stack.

Optical sections were acquired at a digital size of 1024×1024 pixels and averaged 3 times to reduce noise. In all cases, the image obtained was improved (brightness and gamma) and analyzed using the ZEN 2 Blue Edition Software (Carl Zeiss Microscopy GmbH, Göttingen, Germany). In addition, the color for each channel was selected as follows: green for OTR, red for IB4, and blue for CGRP. In all cases, a 2-dimensional projection image and a single-optical section image (where the OTR signal was strong) were imported in Tag Image File Format (TIFF) and were used to compose the multipanel figures. Furthermore, projection images were 3D rendered and a pseudo-3D (2.5D) graphic was made. In addition, an orthogonal projection using 6 to 8 optical sections was performed.

2.2.7. Sodium dodecyl sulfate polyacrylamide gel electrophoresis and Western blot

A group of 12 rats was subdivided into 3 groups (control; saline injection, 50 μL /paw, s.c.; and 1% formalin injection, 50 μL /paw, s.c.; $n = 4$ each) to determine the presence of OTR in the peripheral nerve. After 1 day postinjection, all animals were killed by decapitation and the left sciatic nerve (ipsilateral to s.c. injection) was exposed; the connective tissue was carefully removed and the nerve was cut above the trifurcation of the tibial, common peroneal, and sural rami nerves. Then, the fragments were frozen and stored at -70°C . In all cases, an $\approx 2\text{-cm}$ fragment of sciatic nerve was collected.

The tissue was homogenized and extracted for 2 hours by agitation in ice-cold hypotonic Tris-HCl buffer (Tris 50 mM, pH: 9) containing a protease inhibitor cocktail (Complete Mini; Roche, Mannheim, Germany) and 0.5 mM phenylmethylsulfonyl fluoride (PMSF; Sigma, St. Louis, MO). The homogenates were centrifuged at 10,000 rpm for 20 minutes at 4°C . The supernatants were collected and the protein content was quantified using the Bradford method.⁵ Then 30- μg protein of each sample was resolved by 1-dimensional denaturing sodium dodecyl sulfate polyacrylamide gel electrophoresis in 1.0-mm thick, 6-cm long, 12% gel using the Laemmli³¹ buffer system in a Mini-PROTEAN II cell (Bio-Rad, Hercules, CA). The samples were run under reducing conditions (in the presence of 5% [wt/vol] 2-mercaptoethanol) and constant voltage (100 V stacking gel; 150 V separating gel). Prestained molecular weight markers (Bio-Rad) were used to estimate the molecular weight.

After electrophoresis, the slabs were equilibrated in cold transfer buffer (25 Mm Tris-HCl, 192 mM glycine, 20% methanol vol/vol, pH 8.3) for 30 minutes and electrotransferred (at 200 mA for 1 hour)⁴⁷ to nitrocellulose membranes (Bio-Rad). After transfer, the membranes were washed with 30 mM Tris, 500 mM NaCl Tris Buffered Saline (TBS), at pH 7.5 for 10 minutes, then blocked with 5% (wt/vol) nonfat dry milk (Bio-Rad) in TBS for 2 hours. After washing the membranes 3 times with Tris Buffered saline with tween 20 TTBS (TBS containing 1% [wt/vol] non-fat dry milk, and 0.05% [wt/vol] Tween 20) for 10 minutes, they were incubated overnight at room temperature in TTBS with the OTR antibody (raised in goat, diluted 1:500, Cat.no. sc-8103, Santa Cruz Biotechnology Inc). After that, membranes were rinsed 3 times (10 minutes each) in TTBS and incubated during 2 hours at 4°C with a secondary antibody (rabbit anti-Goat IgG-HRP conjugate, diluted 1:3000, Cat.no. 81-1620; Invitrogen). Protein signal detection was achieved by incubating the membranes in Electrochemiluminescence (ECL) reagent (Amersham-Pharmacia, Buckinghamshire, United Kingdom) for 5 minutes and exposed to Kodak BioMax ML film. The next day, blots were stripped and incubated overnight at room temperature with GAPDH antibody (raised in rabbit, diluted 1:1000, Cat.no. 2118; Cell Signaling Technology); membranes were then rinsed 3 times with TTBS and incubated for 2 hours at 4°C with a secondary antibody (goat anti-Rabbit IgG-HRP conjugate, diluted 1:3000, Cat.no. 65-6120; Invitrogen). The anti-GAPDH was used as internal control to normalize OTR's protein expression level. In both cases, immunoblots were scanned and bands were semiquantitatively analyzed using Image Lab Software (v5.2.1; Bio-Rad Laboratories).

2.2.8. Drugs

In addition to the anesthetic (urethane), this study used the following compounds (obtained from the sources indicated): OT acetate salt hydrate (Sigma Chemical Co, St Louis, MO) and the (S)-2-amino-N-((1S,2S,4R)-7,7-dimethyl-1-((4-*o*-tolylpiperazin-1-yl)sulfonyl) methyl) bicycle [2.2.1] heptan-2-yl)-4-(methylsulfonyl) butanamide hydrochloride (L-368,899 hydrochloride; Tocris Bioscience, Northpoint, United Kingdom). The OT was dissolved in physiological saline, whereas L-368,899 was dissolved in 10% DMSO, and the resulting solution was gauged with saline. These vehicles had no effect on the baseline values of neuronal activity or on behavioral responses. The doses mentioned in this text refer to substance-free base.

2.2.9. Statistical analysis

All data, tables, and figures in the text are presented as mean $\pm$ SEM. The number of evoked potentials induced by electrical stimulation of the paw was normalized and expressed as percentage change from the respective baseline (100%). This baseline response was established after an identified neuron had a $\leq 10\%$ variation in the neuronal responses induced by RF stimulation during 5 consecutive tests (10 minutes between each test). Furthermore, the baseline value refers to evoked neuronal response before peripheral treatment with OT or L-368,899 plus OT. The difference in neuronal activity evoked within 1 group of animals before and after treatment was compared using a 2-way repeated measures analysis of variance. In addition, temporal-course was adjusted to obtain global neuronal activity because of the treatment; in this case, a 1-way analysis of variance was performed. In the formalin test, curves were constructed by plotting the number of flinches as a function of time. The area under the number of flinches against

time curves, an expression of the duration and intensity of the effect, was calculated by the trapezoidal rule, and a 1-way analysis of variance was performed. In the motor coordination test, a Student *t* test was used to compare time to fall. The 1-way and 2-way analyses of variance were followed, if applicable, by the Student–Newman–Keuls post hoc test. In all cases, statistical significance was accepted at $P < 0.05$.

3. Results

3.1. Peripheral local administration of oxytocin inhibits the nociceptive activity of the second-order wide-dynamic-range neurons

Peripheral electrical stimulation of the RF elicited well-defined neuronal responses of the spinal dorsal horn WDR cells (Fig. 1A–D). Briefly, 1 pulse of stimuli in the RF elicited a classical neuronal response of a spinal second-order WDR neuron (Fig. 1B–C). This neuronal activity could be broken down according to the conduction velocities of primary afferent fibers (ie, A β -, A δ -, C fibers, and postdischarge) (Fig. 1D). Interestingly, after a single subcutaneous administration of OT (s.c.; 1–56 μ g/50 μ L) (Fig. 1E–H), a dose-dependent decrease in the firing responses elicited by 20 electrical stimuli on the RF was observed (compared with its respective basal responses). These inhibitions ($\approx$ 40%–60% of basal response) were particularly more pronounced in the A δ fibers, C fibers, and postdischarge, starting 5 minutes after OT administration and lasting up to 100 minutes. When we analyzed these results as global neuronal activity (Fig. 1I–L), we found that OT mainly inhibits the neuronal firing associated with the activation of A δ fibers, C fibers, and postdischarge

(Fig. 1J–L). Figure 1M–P show peri-stimulus time histograms obtained from 1 WDR neuron before and after OT administration; these figures clearly depict the inhibition of C-fiber activity.

3.2. The antinociceptive effect of subcutaneous oxytocin has a local and specific effect

Because the electrophysiological experiments showed that OT selectively inhibits the activity of A δ and C fibers, a set of experiments using the 1% formalin test (50 μ L formalin injected s.c. into the dorsal surface of the hind paw) were performed to test the potential antinociceptive behavioral effects of peripheral local OT. Subcutaneous (s.c.) formalin produced a typical pattern of flinching behavior characterized by a biphasic time course (Fig. 2A). Phase I of the nociceptive response began immediately after formalin injection and then gradually declined ($\approx$ 10 minutes). Phase II began about 15 minutes after formalin injection and lasted for 1 hour. Subcutaneous injection of OT (31 and 100 μ g; ipsilateral to formalin injection) significantly prevented formalin-induced nociception in rats (Fig. 2A). Indeed, OT but not vehicle, prevented flinching behavior in phase I (100 μ g/paw; Fig. 2B) and phase II (31 and 100 μ g/paw; Fig. 2C) of the formalin test. In contrast, contralateral peripheral injection of a supramaximal dose of OT (100 μ g/paw) failed to modify formalin-induced flinching behavior, thus indicating a local effect. This antinociceptive action is not only local but also specific, because the highest dose of OT did not produce any changes in the motor coordination test (Fig. 2D). We also used Western blotting and found that OTR is expressed in the sciatic nerve, which innervates the paw; moreover, the s.c.

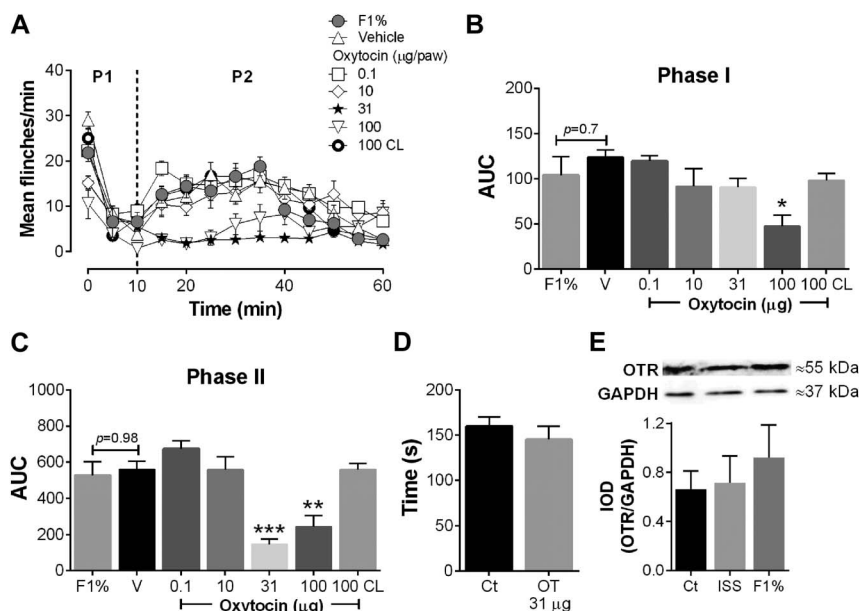


Figure 2. Subdermic oxytocin (OT) administration selectively inhibits behavioral nociception. (A) Time course during phase I (P1) and phase II (P2) of the mean number per minute of flinches observed after subcutaneous (s.c.) treatment with vehicle (V) ($n = 6$) or OT (0.1–100 μ g/paw; $n = 6$ each dose) in rats submitted to the 1% formalin test. Note that although 31 and 100 μ g/paw (administered ipsilaterally to formalin injection) diminish nociceptive behavior, 100 μ g/paw (administered contralaterally [100 CL]; $n = 6$) follows the same pattern as the formalin curve (F1%; $n = 6$) and the control curve (vehicle). Panels B and C show the time course data expressed as area under the mean number of flinches against time curve (AUC). Oxytocin reduced AUC values during phase I (100 μ g/paw) and phase II (31 and 100 μ g/paw), indicating an antinociceptive effect; remarkably, 100 μ g/paw of OT administered contralaterally has no effect on the flinching behavior induced by formalin. * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$, statistically significant difference compared to V. (D) Oxytocin ($n = 5$) has no effect (compared to control; $n = 5$) on the time spent (seconds) on the treadmill apparatus (rota-rod test) before falling, suggesting that OT-induced antinociception is specific. (E) Western blot analysis of the oxytocin receptor obtained from the sciatic nerve of naive rats (Ct; $n = 4$), or isotonic saline solution (ISS; $n = 4$) or formalin (F1%; $n = 4$) injected (50 μ L/paw). The sciatic nerve tissue was collected 1 day after ISS or formalin injection. Data are expressed as the integrated optical intensity. This Figure 2 in A the abscissa has not level and it is TIME. normalized against GAPDH, and they represent the mean $\pm$ SEM of 4 independent experiments.

formalin injection (Fig. 2E) did not seem to modify this expression.

3.3. Blocking the oxytocin receptors abolished the oxytocin-induced antinociception in behavioral and electrophysiological assays

Because OTRs seem to be expressed in the sciatic nerve (Fig. 2E), we examined their functional role in the electrophysiological and behavioral experiments. As shown in Figure 3, a subcutaneous injection of the potent and selective OTR antagonist L-368,899 (10 and 100 $\mu\text{g}/\text{paw}$) was able to block the 31 $\mu\text{g}/\text{paw}$ OT-induced behavioral antinociception (Fig. 3A–C). This effect was clearly observed in phase II of the formalin test (Fig. 3C) and no changes were observed in animals pretreated with the vehicle. Interestingly, an increase in the nocifensive response was elicited in the 10 $\mu\text{g}/\text{paw}$ L-368,899 during phase II (Fig. 3C). Furthermore, when we measured the neuronal activity of a group of WDR neurons (Figs. 3D–H, 4), we found that OT-induced A δ fiber C fiber, and postdischarge inhibition were abolished by the 10 $\mu\text{g}/\text{paw}$ L-368,899, and we observed no effect when we only administered the antagonist (Fig. 4B–D).

3.4. Oxytocin receptors are present in CGRPergic fibers but not in IB4 terminal sensory fibers

Figure 5 shows the immunofluorescence to OTR, IB4, and CGRP from 3 different experiments (A, B, and C); it is important to point out that in the absence of primary antibodies, no unspecific staining was observed (Fig. 5Aiii–Ciii). Furthermore, as shown in Figure 6, OTRs (Ai) do not colocalize with IB4 (Aii) but predominantly colocalizes with CGRPergic fibers (Aiii). In both cases (CGRP+ and IB4+ fibers), the sensory fibers were located between the epidermis and dermis, suggesting that OTR could be expressed in the terminal nerve fibers. It is interesting to note that although the projection image (z-stack projection) (Fig. 6Ai–iv) clearly depicts the presence of OTR, IB4, and CGRP in the tissue, OTRs were mainly colocalized with CGRP as it was observed in 1 representative orthogonal view section (see Fig. 6B). These results are clearly depicted in several representative single optical sections (Fig. 6Ci–iv). Similar results are shown in Figure 7 (see also supplemental Figure 4, available online at <http://links.lww.com/PAIN/A459>), where OTRs (i) did not colocalize with IB4 fibers (ii) but it had a colocalization with CGRPergic fibers (iii).

Finally, although we did not find TB-positive fibers in the periphery, the tracer was successfully carried at the spinal cord (data not shown).

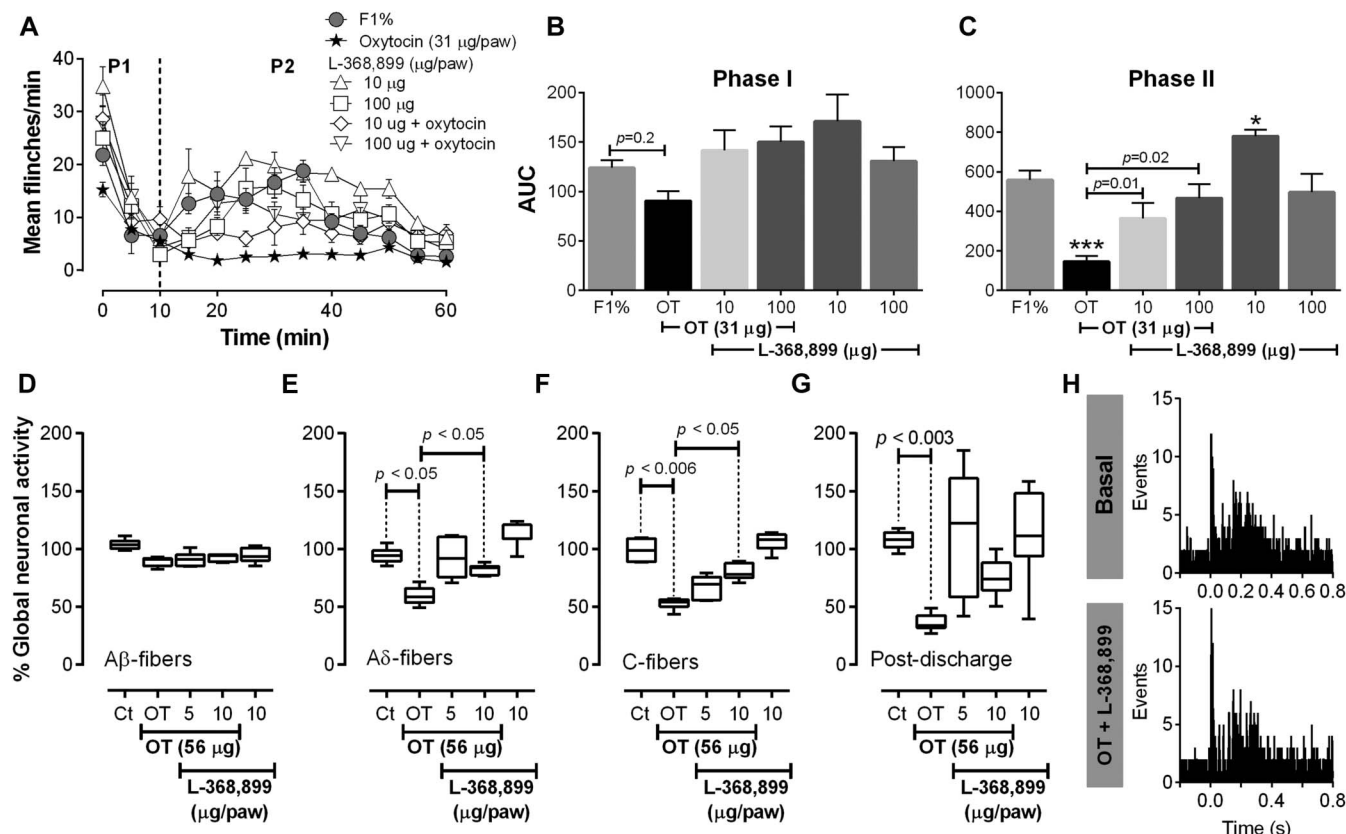


Figure 3. Role of oxytocin receptors (OTRs) in behavioral and electrophysiological oxytocin (OT)-induced antinociception. (A) Time course during phase I (P1) and phase II (P2) of the mean number of flinches observed after subcutaneous treatment with L-368,899 (10 or 100 $\mu\text{g}/\text{paw}$; OTR selective antagonist; $n = 5$ each dose) or L-368,899 plus OT (31 $\mu\text{g}/\text{paw}$; $n = 5$) in rats submitted to the 1% formalin test. Note that L-368,899 reverses the OT-induced antinociception. Panels B and C show the data from the time course expressed as area under the mean number of flinches against time curve. Blockade of OTR by L-368,899 inhibits the OT-induced antinociception in phase II; it is interesting to note that 10 $\mu\text{g}/\text{paw}$ L-368,899 enhances the flinching (C) behavior induced by formalin. $*P < 0.05$ and $***P < 0.001$, statistically significant difference compared to formalin (F1%). (D, E, F, and G) Global neuronal activity (obtained from the respective time course figures, see supplemental digital content, available online at <http://links.lww.com/PAIN/A459>) of the wide dynamic-range (WDR) cells in response to 56 $\mu\text{g}/\text{paw}$ OT ($n = 6$) or 5 and 10 $\mu\text{g}/\text{paw}$ L-368,899 plus OT ($n = 4$ each dose). Pretreatment with L-368,899 was able to block the OT-induced inhibition of neuronal activity associated with A δ -fiber and C-fiber activation. Note that 10 $\mu\text{g}/\text{paw}$ L-368,899 alone ($n = 4$) has no significant effect on the neuronal-evoked responses. Panel H shows 2 PSTHs obtained before (basal response) and after OT + L-368,899 (1 hours after oxytocin) for 1 WDR neuron, thus illustrating that L-368,899 (56 $\mu\text{g}/\text{paw}$) inhibits the OT-induced antinociception (compared with Fig. 1N–P).

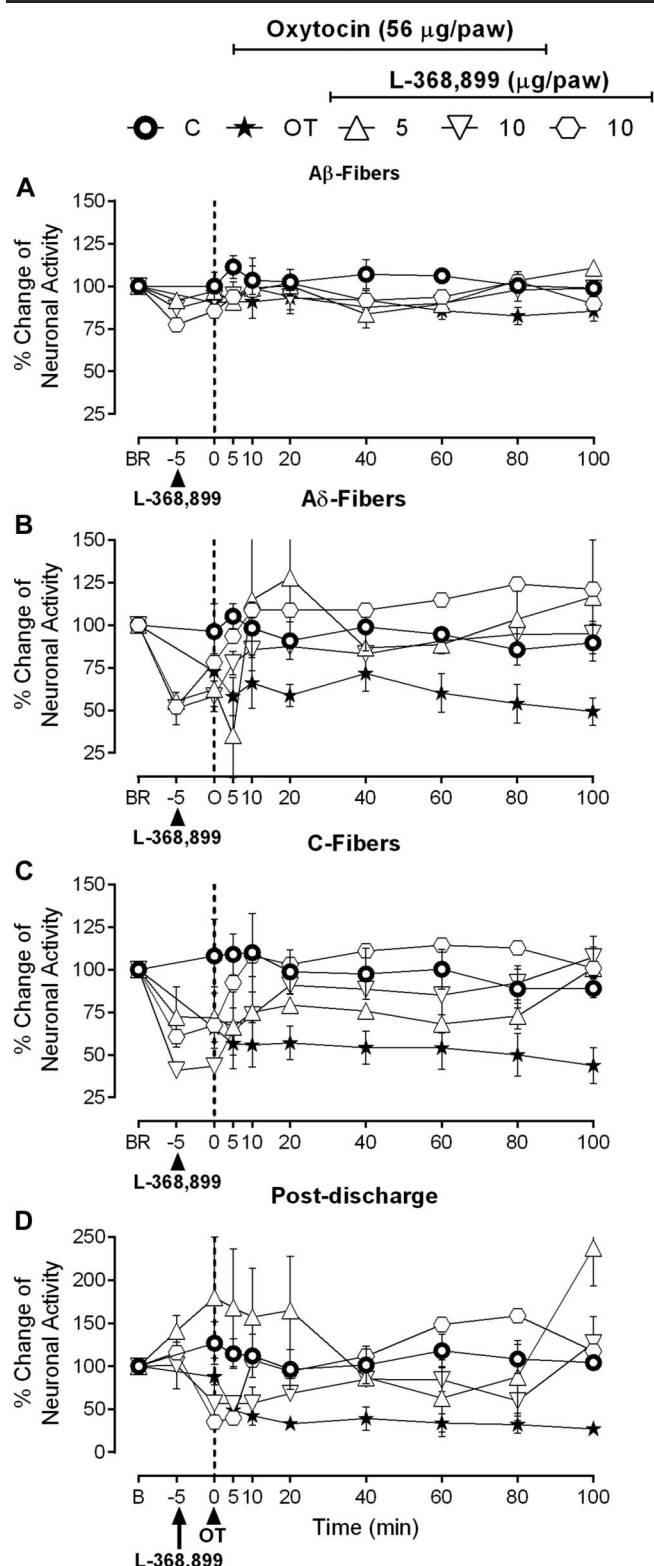


Figure 4. Blockade of oxytocin receptors (OTRs) inhibits the oxytocin (OT)-induced antinociception in the second-order wide-dynamic-range (WDR) cells mediated by (A) A β -, (B) A δ -, (C) C fibers, and (D) postdischarge. Effect of subcutaneous bolus injection of L-368,899 per se or on the inhibition of nociceptive WDR responses induced by OT (56 μ g/paw) of the receptive field electrical stimulation. * $P < 0.05$ vs the corresponding control response (●). BR, basal response, before any administration.

4. Discussion

4.1. General

Our findings show that peripheral activation of OTRs on the superficial skin layer is able to induce inhibition of the C-fiber discharge and suppress behavioral nociception. Oxytocin receptors could be expressed in CGRPergic terminal nerves innervating the superficial skin. The role of this receptor is supported considering that subcutaneous OT-induced antinociception was reverted by L-368,899 (a potent and selective OTR antagonist) in electrophysiological and behavioral experiments. This effect seems to be local, because OT ipsilateral (not contralateral) to formalin inhibited nociception in the formalin test.

4.2. Subcutaneous oxytocin blocks the nociceptive input to the spinal dorsal horn wide-dynamic-range cells

As previously reported,⁵⁰ electrical stimulation of the peripheral RF produces a typical well-defined triphasic neuronal-evoked response corresponding to A β -, A δ -, and C-fiber activation. We found that OT administered s.c. into the RF inhibits the evoked nociceptive (A δ fiber and C fiber) neuronal responses (Fig. 1) and this effect lasts up to 100 minutes. Although at this point we cannot ascertain the specific location of OT effects (ie, peripheral vs central), some studies suggest that OT not only induces analgesia in the spinal cord of rodents^{8,37,57} and humans,^{9,10} but also inhibits nociception at the peripheral level by acting on OTRs located in the DRG,^{16,26,30} as described in the Introduction section.

Indirect evidence for this hypothesis came from an early study in a mouse model of chronic abdominal pain where an in vivo intracolonic administration of a selenoether OT analog with similar biological activity in OTR than OT was able to inhibit chronic visceral hypersensitivity.¹¹ This suggestion gains weight when considering the pharmacokinetic properties of several neuropeptides, including OT (such as poor absorption and low metabolic stability).²⁹ Therefore, it is reasonable to suppose that the pharmacological effect observed after single s.c. OT administration is restricted to a local effect. To validate this hypothesis, we performed behavioral, molecular, and pharmacological tests.

4.3. Oxytocin inhibits the formalin-induced flinching behavior by a local and specific effect

Because s.c. OT inhibits the nociceptive-evoked activity of WDR cells, we hypothesized that this inhibition should be replicated in a pain behavior model. In accordance, flinching behavior induced by s.c. formalin is dose-dependent reduced by OT in both phases (Fig. 2A–C). Considering that 100 μ g/paw OT in the contralateral paw was ineffective, this neuropeptide seems to be a local effect. Moreover, although 100 μ g/paw OT inhibited the nociceptive behavior in both phases, it has been reported that 200 to 1000 μ g/kg OT (50–250 μ g for a 0.25 kg rat) administered s.c. inhibited the locomotor activity (ambulatory and nonambulatory) by activating peripheral V1a receptors.⁵⁴ To avoid possible bias because locomotor performance could influence the nociceptive tests,²⁷ we decided to use the dose of 31 μ g/paw OT. In any case, the 31 μ g/paw (that inhibited phase II of the formalin test) did not have any effect on the motor coordination test when using the rota-rod test (Fig. 2D). The notion that s.c. OT uses a local mechanism to exert its antinociceptive effect is strongly supported in conjunction with the fact that OTRs are expressed in the sciatic nerve (Fig. 2E).

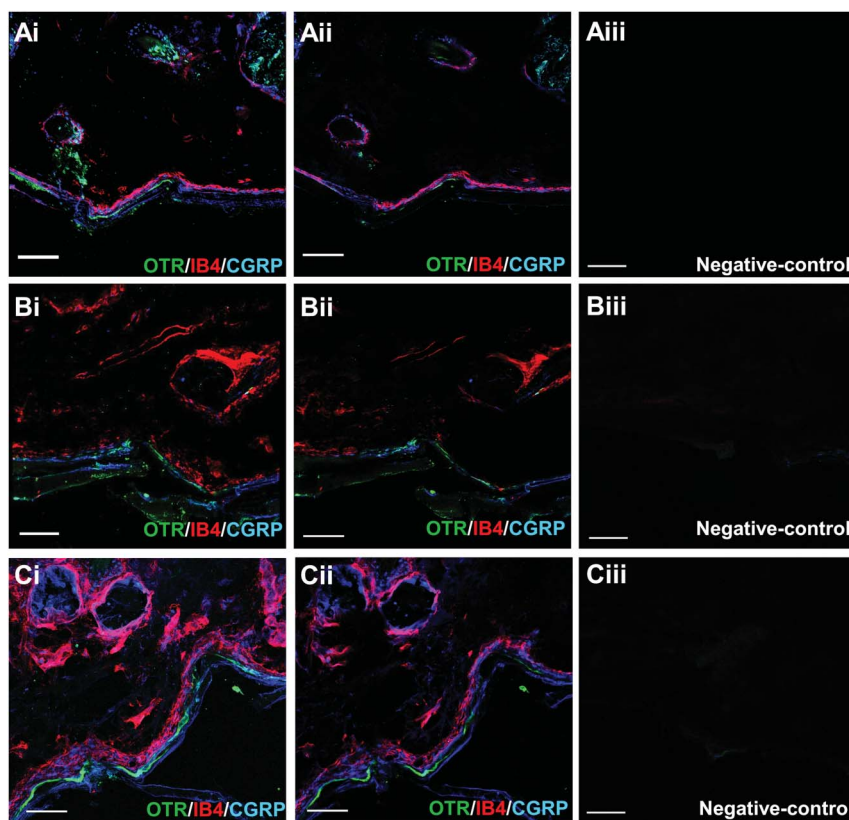


Figure 5. Confocal image of a skin section for oxytocin receptor (OTR), IB4, and calcitonin gene-related peptide (CGRP) immunofluorescence. Panels Ai, Bi, and Ci are the z-stack projection from 40 optical sections (25×/0.8 NA objectives) from 3 different experiments. Panels Aii, Bii, and Cii are single optical sections from the z-stack projection; the optical section was selected at the point that signal for OTR was better (see supplemental Figures 1, 2, and 3 for details about the OTR colocalization with CGRP and IB4; available online at <http://links.lww.com/PAIN/A459>). The co-localization between OTR and CGRP is shown in white-grey colors. Panels Aiii, Biii, and Ciii are the negative controls showing no specific labeling in the absence of primary antibodies. It is important to point out that in this set of experiments the intensity color was adjusted at the maximum to appreciate possible unspecificity from the secondary antibodies used. Scale bars: 50 μ m.

4.4. Defining the role of peripheral oxytocin receptors in peripheral oxytocin-induced antinociception

Direct evidence of the involvement of peripheral OTRs in OT-induced antinociception in behavioral (Fig. 3A–C) and electrophysiological (Fig. 3D–H) experiments originates from the fact that antinociception was clearly antagonized by L-368,899, a specific and potent OTR antagonist⁵³ (Fig. 4). Correspondingly, we found that OTRs could be localized in cutaneous nociceptive peptidergic terminals (Figs. 5–7). Previous reports have strongly supported the role of peripheral OTRs in nociception,^{16,26} but the potential role in peripheral nociceptor endings remains obscure, as previously suggested.¹⁹ In this context, a previous study using an isolated ex vivo preparation showed that OT could block the capsaicin-induced release of CGRP from dural nociceptors,⁴⁹ supporting the idea that OTRs could be present in terminal nerve endings. Certainly, apart from the spinal cord^{40,48,55} DRG⁴⁰ and trigeminal ganglia,⁴⁹ OTR was found in mouse embryonic skin²¹ supporting our hypothesis. In this context, our work is the first to show that OTRs can be found in nociceptive terminals and that on nociceptive stimulation they are able to inhibit the nociceptive input.

4.5. Considerations about the specificity of the antibody against oxytocin receptor

During the submission process of the present paper, the reviewers strongly suggested to perform an immunostaining control to avoid the possible false positive OTR in skin terminals. In Figure 5, the

first antibody missing showed nonunspecific staining. In addition, during the submission, a new possibility of OT action was published in PAIN in which Kubo et al.³⁰ suggested that the OT effects suppressing orofacial hypersensitivity can be mediated by the vasopressin (VP)-1A receptor (V_{1A}-R). This assumption is based on the action of a selective V_{1A}-R antagonist over isolated trigeminal ganglia neurons. Considering the molecular structure similarity between OT and VP, this assumption is a possibility as was also suggested elsewhere⁴⁴ using OT knockout mice. However, the isolated trigeminal ganglia neurons³⁰ as well as the knockout OT mice⁴⁴ used could develop compensatory mechanisms to compensate the lack of OT regulations. Nevertheless, the question that remains is whether OT could have a cross-action with the V_{1A}R, although this question does not invalidate that our current description of peripheral OTR used OT to act. Considering the molecular, electrophysiological, pharmacological, and behavioral results and that the current OTR has a juxtaposition with peptidergic CGRP fibers, an important possibility is that OT uses the OT receptor.

A serious characteristic for using antibodies in research is their specificity, a property dependent on the species studied. In a recent study, using OTR knockout mice⁵⁶ has suggested that the antibody used in our study has a lack of specificity to detect adequately the presence of OTR. However, the large number of regulatory elements controlling transcription and tissue-specific localization of the OTR¹⁷ and the fact that knockout mice could develop an array of compensatory

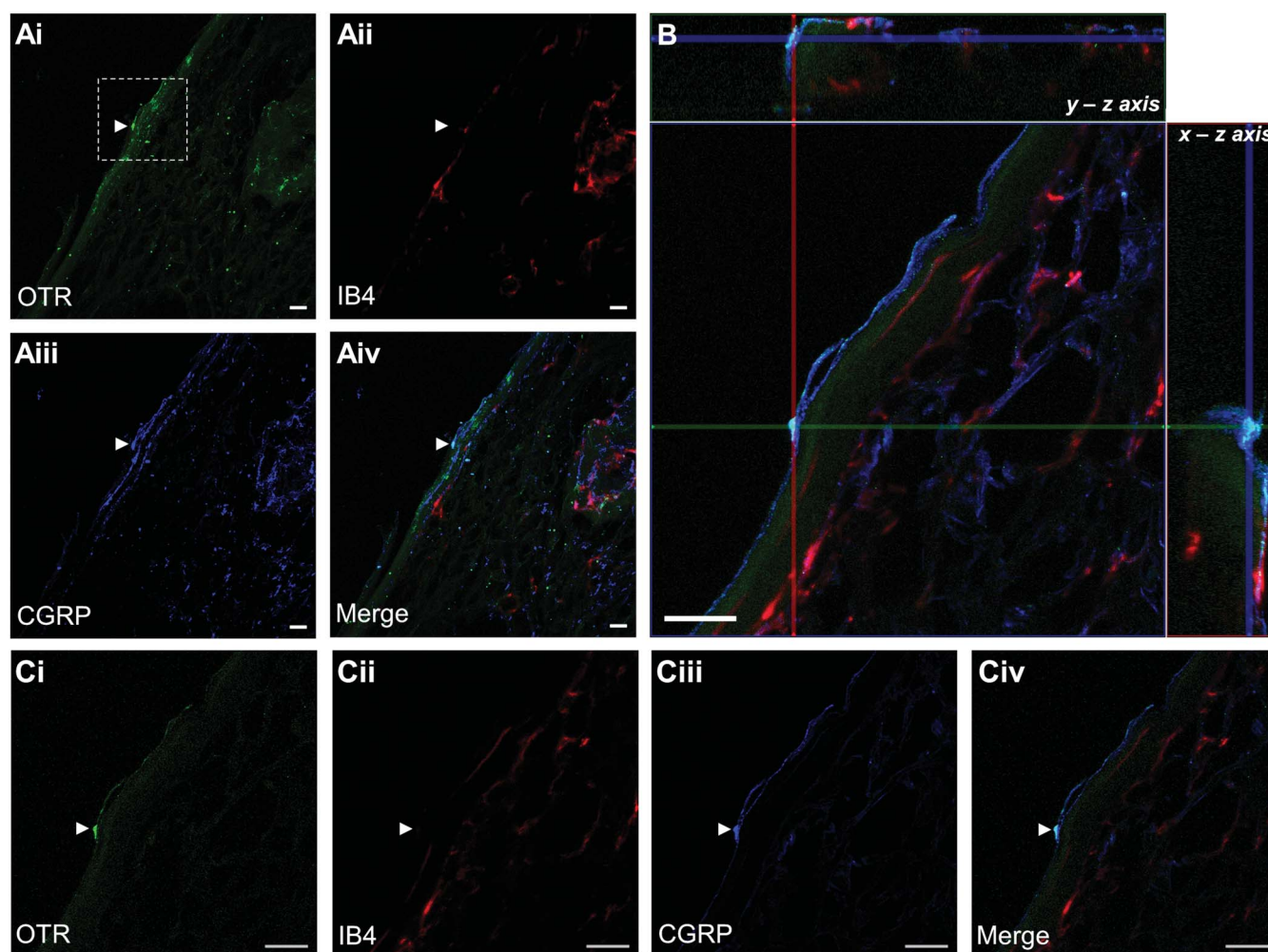


Figure 6. Triple labeling of terminal nerve endings in dermis and epidermis sections with antibodies against oxytocin receptor (OTR), calcitonin gene-related peptide (CGRP), and IB4. Confocal microscopic images of OTR (green), IB4 (red), and CGRP (blue) immunofluorescence in the skin. Ai, ii, iii, and iv are z-stack projection from 8 optical sections; images were taken with $25\times/0.8$ NA objectives. The arrows show the free termination of OTR positive. Panel B shows an orthogonal view displaying 8 optical sections ($1\ \mu\text{m}$ each) at $40\times/1.3$ NA; y-z plane and x-z axis show colocalization between OTR (green) and CGRP (blue) but not with IB4 (red). The co-localization between OTR and CGRP is shown in white colors. Ci-Civ show a focal plane of the z-stack; OTR and CGRP appear in the same plane and the arrow shows the free termination OTR and CGRP positive; this image was taken with $25\times/0.8$ NA objectives. Scale bars: $50\ \mu\text{m}$.

mechanisms to adjust the absence of deleted receptors (ie change the endogenous protein expression profile²³), this transgenic approach could lead to a potential flaw in the immunodetection of this receptor. Admittedly, the evidence presented by Yoshida et al.⁵⁶ could undermine our findings using immunofluorescence, but we need to keep in mind that perhaps the knockout OTR model is not the best model to evaluate the antibody specificity for a key receptor involved in several significant physiological functions.¹⁸ Thus, the physical presence of OTR at the periphery will be definitely established with the advent of more selective OTR antibodies, which to the best of our knowledge, are not currently available. In any case, our pharmacological experiments, (in electrophysiological and behavioral tests) using a highly potent and selective OTR antagonist, support the presence of this receptor at the peripheral level. Consequently, it is important to point out some key aspects to properly appreciate the results found: First, in our manuscript we discuss and present evidence of others research groups suggesting that OTR could be expressed in the peripheral nerves.^{11,13,14,16,21,26,49}

Second, the antibody used is recommended for the detection of OTR by Western blotting, immunofluorescence, and

enzyme-linked immunospecific assay. Apart from the use of mice, the work done by Yoshida et al.⁵⁶ was performed using immunohistochemistry. Although immunohistochemistry and immunofluorescence are similar techniques, a key information in the work of Yoshida et al.⁵⁶ is missing (the dilution used). Indeed, the antibody concentration used is critical to avoid false positive results when considering that VP AV1a receptors are similar to OTR and share a high degree of homology.^{28,29} Furthermore, the “un-specific” antibody (LS-A246) tested by Yoshida et al.⁵⁶ (antibody to OTR) was primarily designed to react preferentially with the human OTR instead of the rodent OTR. Indeed, the LS-A246 was raised against synthetic 16 amino acids from the third cytoplasmic domain of the human OTR. This is relevant considering that although a high degree of homology between rat and human OTR exists, the third intracellular loop are different between these 2 species,^{1,28,39} and maybe the fact that the signal by LS-A246 was observed in knockout mice⁵⁶ reflects the complexity of compensatory mechanisms in these animals.

Subsequently, the specificity and sensitivity of the antibody that we used has been tested by several research groups analyzing not only the role of OTR in nociception³⁸ but also in autonomic and cardiovascular systems.^{6,20,24,32,45,51} Hence, the interesting

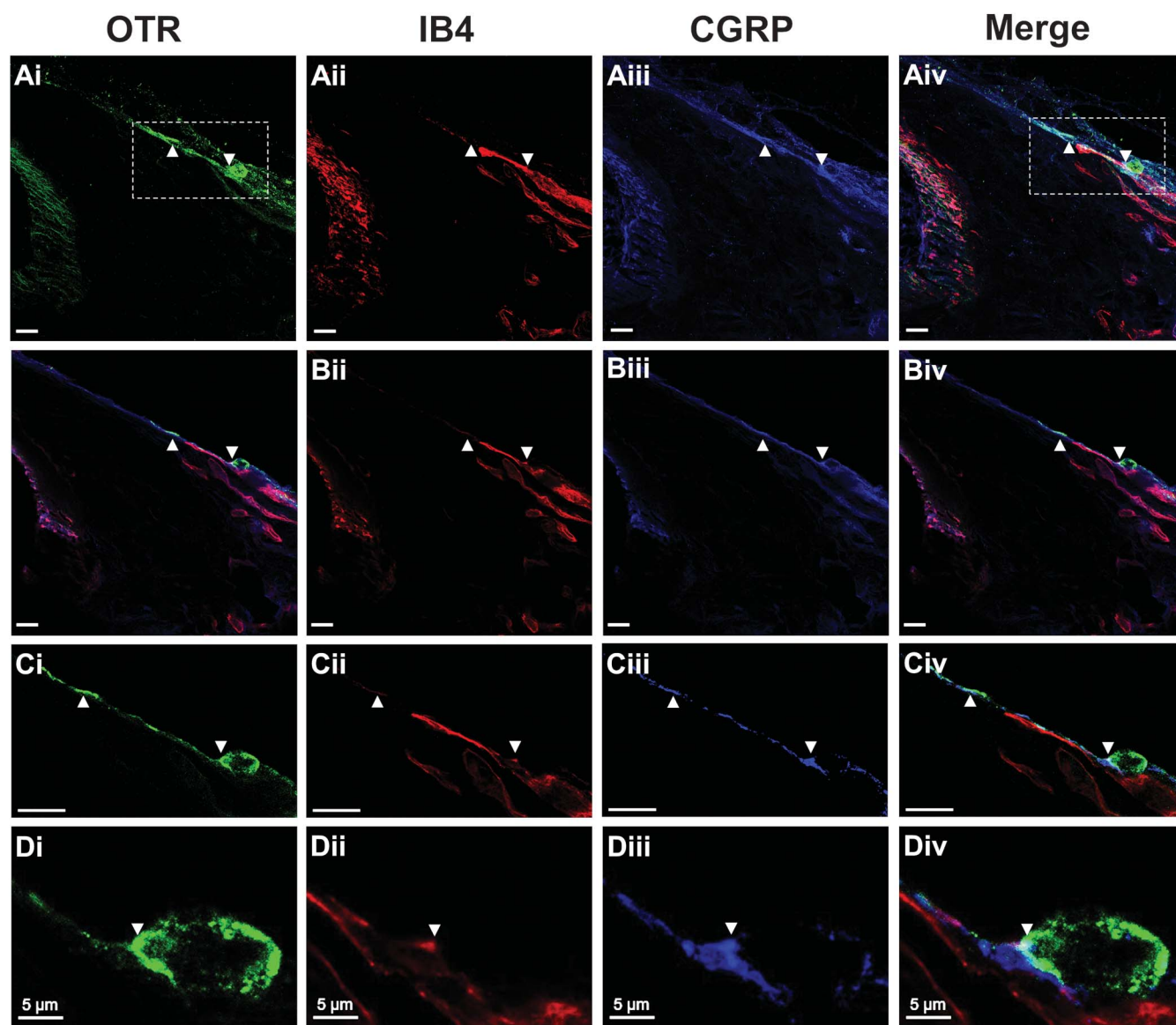


Figure 7. Triple-labeling of terminal nerve endings on dermis and epidermis sections with antibodies against oxytocin receptor (OTR), calcitonin gene-related peptide (CGRP), and IB4. Representative confocal images of immunofluorescent-labeled terminal sensory nerves in the lateral hairy skin paw (OTR in green, IB4 in red, and CGRP in blue). Panels in A are the z-stack projection from 40 optical sections; the images were taken with 25×/0.8 NA objectives. The arrows show the OTR-positive fiber and cellular body. Panels in B are single optical sections taken with 40× objective (section thickness 1 μ m) from the z-stack projection in A. Panels in C are single optical sections (40×/1.3 NA) from the box in panel A. Panels in D are cropped images from panel C showing a terminal nerve ending. Although the z-stack projection (Ai, ii, iii, and iv) depicts the presence of OTR, IB4, and CGRP in the tissue, OTRs are predominantly colocalized with CGRP when we observe the 1 representative optical section (see B, C and D). The co-localization between OTR and CGRP is shown in white colors (see merge). These results are clearly depicted in several representative optical sections (orthogonal view, supplemental Figure 4, available online at <http://links.lww.com/PAIN/A459>), and this suggestion is reinforced by similar data obtained from the 3D-rendered and 2.5D-reconstructed images (see supplemental Figures 5 and 6, available online at <http://links.lww.com/PAIN/A460> and <http://links.lww.com/PAIN/A461> respectively). Scale bar in panels A, B, and C: 20 μ m.

findings by Yoshida et al.⁵⁶ suggesting that this antibody is unspecific need to be taken cautiously.

Finally, regarding the physiological function of OTRs on peripheral skin, OT is expressed in cultured human keratinocytes and is released in response to external stimuli (resembling injuries).¹⁴ Skin keratinocytes responding to stimulation can indirectly modulate the activity of sensory fibers¹³ and could modulate action potentials in primary afferent fibers.⁴

5. Conclusion

Our work suggests that the antinociception induced by subcutaneous OT is mediated by a peripheral mechanism (probably

at the terminal nociceptive fiber endings by OTR activation), and it reveals a new potential role for OT in pain modulation.

Conflict of interest statement

The authors have no conflicts of interest to declare.

Acknowledgments

The authors thank Jessica González Norris for reviewing the grammatical aspects of this paper. In addition, the authors also acknowledge Elsa Nydia Hernández Ríos for her technical assistance in confocal microscopy. This work was sponsored by

grant (to MCL) PAPIIT-UNAM (Grant No. IN200415). IATG is a CONACyT Fellow. Alfredo Manzano García is a doctoral student from Programa de Doctorado en Ciencias Biomédicas, Universidad Nacional Autónoma de México (UNAM) and received fellowship 597467 from CONACyT. A. González-Hernández and A. Manzano-García contributed equally to the work.

Appendix A. Supplemental digital content

Supplemental digital content associated with this article can be found online at <http://links.lww.com/PAIN/A459>.

Supplemental video content

Video content associated with this article can be found online at <http://links.lww.com/PAIN/A460>, <http://links.lww.com/PAIN/A461>, and <http://links.lww.com/PAIN/A462>.

Article history:

Received 2 December 2016

Received in revised form 4 May 2017

Accepted 30 May 2017

Available online 19 July 2017

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